

Education

University Of Maryland, College Park

College Park, MD

Computer Science, Undergraduate Student

Aug. 2022 - May. 2026

- **Relevant Coursework:** Introduction to Algorithms, Design and Analysis of Computer Algorithms, Introduction to Machine Learning, Introduction to Artificial Intelligence, Introduction to Computer Systems, Advanced Data Structures, Applied Probability and Statistics, Discrete Structures, Linear Algebra, Operating Systems

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, HTML, CSS, Rust

Developer Tools: GitHub, Git, Google Cloud, Firebase, ClearCase, Jira, PyCharm, VS Code, Docker

Technologies/Frameworks: PyTorch, Pandas, SQL, Numpy, Jupyter Notebook, UNIX/Linux, React JS, Tauri

Concepts: REST APIs, CI/CD, Microservices, Linux System Administration, WebAssembly

Experience

Yellow.ai

Remote / Charlotte, NC

AI Engineering Intern

Aug 2025 - Sept 2025

- Designed and prototyped an AI-powered content moderation tool capable of classifying photos and videos as "appropriate" or "inappropriate," with customizable thresholds defined by users.
- Leveraged machine learning libraries such as scikit-learn to build and train models for multimedia classification, integrating them into a scalable prototype for CDN filtering.
- Collaborated with product teams to align technical implementation with customer experience goals, ensuring flexibility and adaptability of the moderation system.

Huntington Ingalls Industries - Mission Technologies Division

Hanover, MD

Summer Intern

June 2025 - Aug. 2025

- Engineered C++ atop log parser and visualizer for Minotaur systems with vanilla HTML/CSS/JS frontend to analyze crash diagnostics from gigabyte-scale datasets in restricted deployment environment.
- Optimized thread management and memory leak detection by parsing RAM trend patterns to identify unstable microservices during high-load stress testing while maintaining zero-internet-access security protocols.
- Explored fault prediction using ML on historical log data for early failure identification.
- Developed efficient chunked file parsing with tree-based lookups to extract crash-causing events from atop logs.

Huntington Ingalls Industries - Mission Technologies Division

Hanover, MD

Summer Intern

June 2024 - Aug. 2024

- Contributed to the development of Minotaur, a specialized platform integrating sensors, cameras, radar, and communication equipment into a unified system for Air and Marine Operations.
- Enhanced software functionality of Minotaur by implementing validation checks, such as ensuring user inputs do not exceed critical thresholds, significantly improving system stability and user experience.
- Expanded Minotaur's capabilities by designing and integrating a new function to filter signals based on specific properties, addressing critical gaps in signal processing functionalities.
- Applied expertise in parsing and analyzing system performance data using atop to optimize software performance and troubleshoot operational issues effectively.

Qure.AI

Charlotte, NC

Summer Intern

May 2021 - Aug. 2021

- Studied convolutional neural networks, AI image recognition, and open-source machine learning architectures to build an AI-based image recognition program to detect medical abnormalities, such as the impact of COVID-19 on the lungs, captured in x-ray and MRI images, as well as size, location and growth rates of brain tumors.
- Gathered, arranged, and corrected research data of over 30,000 x-ray and MRI images, to create representative graphs and charts highlighting results for presentations.
- Implemented advanced Machine Learning concepts such as back-propagation and rectified linear activation functions, which both increase the accuracy and efficiency of the AI model, thereby achieving a 98% detection accuracy rate for the final model.
- Implemented an interactive UI for medical professionals to input multiple images at a time and quickly get results.

Personal Projects

PureMark

Ongoing Development

- Cross-platform Markdown editor with live PDF preview, enabling professional document editing locally without requiring a system LaTeX installation.
- Engineered a Rust backend (Tauri 2) that bundles Pandoc and Tectonic, allowing seamless PDF/DOCX/LaTeX export on Linux, Windows, and macOS.
- Optimized performance through debounced re-renders and lazy rendering.
- Solved platform-specific challenges like WebKitGTK security taints and Pandoc binary path resolution by implementing dynamic shimming and cross-platform build logic.

GridDrop

Ongoing Development

- Created a private local network file transfer utility for direct file sharing between Android iPhone devices with zero apps, setup, and internet required on receiving device.
- Implemented resumable chunk transfers, enabling file uploads/downloads survive flaky connections without restart.
- Solved Android Wi-Fi hotspot API limitations for cross-platform compatibility between older/non-Pro devices and newer phone.

Ryujinx Configurator

Ongoing Development

- Created a controller-navigable utility for Steam Deck/Linux, enabling in-gamepad binding directly from Steam Big Picture Mode without leaving gaming context.
- Reverse-engineered a non-standard controller GUID format used by the Ryujinx emulator, implementing byte-order conversion to ensure compatibility across hardware and streaming platforms.
- Designed a state-machine UI driven by SDL input, with atomic file writes and automatic backup logic to safely manage emulator configuration without corruption risk.
- Built a diagnostic overlay for debugging controller mismatches, improving setup time for couch co-op sessions from minutes to seconds.

PhishNet AI

April 2025

- Chrome web extension protecting users from phishing and malicious websites, integrating VirusTotal API and Gemini AI for threat explanation.
- Implemented automated domain blocking to prevent access to identified threats, reducing security risk during browsing sessions.
- Overcame Chrome Manifest V3 restrictions by designing secure API routing workarounds for real-time threat detection without breaking browser security.